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Summary:

The process of intake of oxygen rich air and giving out of air rich in carbon dioxide is generally called respiration. Pollutants and micro organism are filtered from the inspired air by the hair and mucus present in the nostrils. The two main steps in the mechanism of respiration are inspiration and expiration which takes place due to pressure gradient in the atmosphere and lungs.

O_2 is transported in blood in dissolved form and is also bound to haemoglobin. One molecule of haemoglobin can bind four molecules of O_2. The Sigmoid shape of the O_2 haemoglobin dissociative curve shows increased affinity for each O_2 molecule.

CO_2 is transported in blood in dissolved form as carbamino haemoglobin and as HCO_3. HCO_3 is produced in RBCs from CO_2 and water catalysed by carbonic anhydrase. Breathing is controlled by medullary respiratory centre.

Respiratory volumes and capacities indicate the amount of air inspired and expired during normal respiration. Our respiratory system can be affected by pollutants, pathogens and other chemical substances found in air. Lung cancer and emphysema cannot be cured and these diseases are common among cigarette smokers.

People at higher level than the sea level are prompted to altitude sickness as the barometric pressure is low in those regions. Surfactant, emphysema, Asthma and Dead space have been discussed. During vigorous exercise the rate of respiration increases.

Glossary

Apnoea – Temporary stopping of respiration.

Book gills – Respiratory organs in aquatic Limulus.

Book lungs – Respiratory organs of Scorpions and most spiders.

COLD – Chronic Obstructive Lung Disease.

Dyspnoea – painful respiration.

Epiglottis – a thin elastic cartilaginous flap which covers the glottis and prevents the entry of food into the larynx.

Haemoglobin – iron containing red pigment of RBCs of vertebrates, gives red colour to blood.

Herring-Breuer reflex – a defensive mechanism against over dilation of lungs.

Hypoxia – the failure of tissues for any reason to receive an adequate supply of oxygen.

Pneumothorax – presence of air in the pleural cavity which causes collapsing of lungs.

Vocal cords – sound regulating cords also called larynx or voice box.

Yawning – prolonged inspiration due to increase in CO_2 concentration.